Unified Forensic Investigation Report

Case: Demo: northwind.com phishing intrusion (seed 42) **Generated:** 2026-10-30T13:09:51Z **Generator:** template

Statement verification: 43 of 43 sentences verified against cited evidence (traceability 100.0%).

Evidence inventory

ID	File	Module	SHA-256	Integrity
E1	01_phishing.eml	email	`131b6ce65a66d3ca0ce4b22300fa642dc95d70ac6965f1a9f9fd983f3e1fdb7a`	OK
E2	02_victim_History	browser	`c36240a27bcc2811299f1c57f9ba0fa284377e3f178e594b1ecc70b005f527d4`	OK
E3	03_victim_capture.pcap	network	`cec137a294ec8cc4127e73fb3ceb0db2a4aced09e982b62efde4c3d0f192db41`	OK
E4	04_victim_memory.mem	memory	`5307ace7f5d487a4e0e2943022a3efe2303c28f700e30a555df311af644af17b`	OK
E5	05_honeypot_cowrie.json	deception	`7830e7e2103b4b138003496b3aee066e10c1f60b9be08ef9aff74cf28e6d9e68`	OK
E6	06_victim_disk.body	disk	`4a246eaa73687fd1994b2a222ae5aa6ae265b40e7e8d701af4d711ea8bece62d`	OK
E7	07_receipt_5425.jpg	media	`8e4ff66b53006f0208848516de97c961578adf60198ee8da7b6bb6f462e808c2`	OK
E8	08_newsletter.eml	email	`bfc6385bbc02226317d05da25d839c7a84292d7da1324779c19ca42144435bca`	OK
E9	09_internal_memo.eml	email	`331a1c351f399a5233d9d3062b0b3af335b9108ad65310f277b3e254aaa7209a`	OK
E10	10_scanner_cowrie.json	deception	`8ada32f054c69334067faf210cdf4319efd7ad5dee8afe4771df92d3e961774e`	OK
E11	11_holiday_558.jpg	media	`49e68e2cb3770c01365f727214ddb122637e55601247e4bc1d6745305078b1eb`	OK
E12	12_colleague_History	browser	`5c9de308a737e2b7ef9b3f8589341a1d3c8c75e28051b6f65a112692cbe3f6e6`	OK
E13	13_colleague_capture.pcap	network	`b83279cac76f3e6a969482784761bc32b7337350174f0a54b824f2c4588ec6af`	OK
E14	14_voicemail.wav	media	`5492945241eab610ec92fb2742d0106e73cbf6a7746145d960314af1cee2f1f8`	OK
E15	15_cctv_clip.mp4	media	`4cbd925c2c5b0219ddc401181d8a2af24565511d06397f7ea5b2d65489bcdf79`	OK

Executive summary

• 15 evidence items from the browser, deception, disk, email, media, memory, network modules were analysed, and 15 of 15 matched their acquisition SHA-256 hash on re-verification [E1:131b6ce6][E2:c36240a2][E3:cec137a2][E4:5307ace7][E5:7830e7e2][E6:4a246eaa][E7:8e4ff66b][E8:bfc6385b][E9:331a1c35][E10:8ada32f0][E11:49e68e2c][E12:5c9de308][E13:b83279ca][E14:54929452][E15:4cbd925c].

• Correlated incident 1 links 6 items across browser, deception, disk, email, memory, network evidence through domain hub-cloud72.ru, url http://hub-cloud72.ru/dl/po_384.docm, md5 8eded35883138246572a57a382cf77f1 [E1:131b6ce6][E2:c36240a2][E3:cec137a2][E4:5307ace7][E5:7830e7e2][E6:4a246eaa].

• Decoy honeytokens were found in 2 evidence item(s), which indicates attacker contact with planted bait [E4:5307ace7][E5:7830e7e2].

Attack chain (MITRE ATT&CK mapping)

• Initial Access: T1566 Phishing, T1566.001 Spearphishing Attachment, T1566.002 Spearphishing Link (4 artifact(s), first seen at 2026-08-23 09:27 in 01_phishing.eml) [E1:131b6ce6][E2:c36240a2].

• Execution: T1204 User Execution (1 artifact(s), first seen at 2026-08-23 09:48 in 02_victim_History) [E2:c36240a2].

• Credential Access: T1078 Valid Accounts (decoy credential) (1 artifact(s), first seen at 2026-08-26 11:14 in 05_honeypot_cowrie.json) [E5:7830e7e2].

• Discovery: T1082 System Information Discovery (3 artifact(s), first seen at 2026-08-26 11:14 in 05_honeypot_cowrie.json) [E5:7830e7e2].

• Command and Control: T1071 Application Layer Protocol (3 artifact(s), first seen at 2026-08-23 09:43 in 03_victim_capture.pcap) [E3:cec137a2][E4:5307ace7].

• Exfiltration: T1071.004 DNS (8 artifact(s), first seen at 2026-08-23 09:58 in 03_victim_capture.pcap) [E3:cec137a2].

Key findings by evidence item

• Email evidence 01_phishing.eml: Email: Action required: password will expire at 2026-08-23 09:27 (SPF result is 'fail'; DKIM result is 'none') [E1:131b6ce6].

• Email evidence 01_phishing.eml: Attachment po_384.docm at 2026-08-23 09:27 (Risky attachment type: po_384.docm) [E1:131b6ce6].

• Browser evidence 02_victim_History: Downloaded po_384.docm at 2026-08-23 09:48 (Downloaded risky file type po_384.docm; Suspicious TLD .ru (hub-cloud72.ru)) [E2:c36240a2].

• Browser evidence 02_victim_History: Visited https://northwind-verify-portal.click/verify?id=65392 at 2026-08-23 09:46 (Suspicious TLD .click (northwind-verify-portal.click); Credential-themed URL on hyphenated look-alike host northwind-verify-portal.click) [E2:c36240a2].

• Browser evidence 02_victim_History: Visited http://hub-cloud72.ru/dl/po_384.docm at 2026-08-23 09:48 (Suspicious TLD .ru (hub-cloud72.ru)) [E2:c36240a2].

• Network evidence 03_victim_capture.pcap: UDP 10.8.52.183 → 1.1.1.1:53 at 2026-08-23 09:43 (Beaconing: 11 sessions to 1.1.1.1:53 every ~7s) [E3:cec137a2].

• Network evidence 03_victim_capture.pcap: TCP 10.8.52.183 → 193.214.112.115:8443 at 2026-08-23 09:50 (Beaconing: 8 sessions to 193.214.112.115:8443 every ~60s) [E3:cec137a2].

• Network evidence 03_victim_capture.pcap: DNS d7dmxmhkhtqxsy7elgs4gtj3w2cxpjt5fwi3w4qtu364u2qe.t.hub-cloud72.ru at 2026-08-23 09:58 (Possible DNS tunnelling: long/high-entropy label (48 chars, H=4.51)) [E3:cec137a2].

• Network evidence 03_victim_capture.pcap: DNS 4ucvgs54rk5dqbwvqzty35qywfbt6sj6xocggtvsjd42tyeu.t.hub-cloud72.ru at 2026-08-23 09:58 (Possible DNS tunnelling: long/high-entropy label (48 chars, H=4.43)) [E3:cec137a2].

• Memory evidence 04_victim_memory.mem: Implant configuration block (synthetic test marker) at offset 562504 (Implant configuration block (synthetic test marker)) [E4:5307ace7].

• Deception evidence 05_honeypot_cowrie.json: Honeypot login SUCCESS svc_sql78/duZEitXA8c6hGd from 23.15.47.56 at 2026-08-26 11:14 (Attacker authenticated to decoy) [E5:7830e7e2].

• Deception evidence 05_honeypot_cowrie.json: Honeypot connection from 23.15.47.56:45241 at 2026-08-26 11:14 (Interaction with decoy service (no legitimate use)) [E5:7830e7e2].

• Deception evidence 05_honeypot_cowrie.json: Honeypot command: uptime at 2026-08-26 11:14 (Command executed in decoy shell) [E5:7830e7e2].

• Deception evidence 05_honeypot_cowrie.json: Honeypot command: ls -la at 2026-08-26 11:14 (Command executed in decoy shell) [E5:7830e7e2].

• Media evidence 07_receipt_5425.jpg: Image 07_receipt_5425.jpg: likely manipulated (score 0.87) at 2026-07-18 13:25 (Localised ELA anomaly (robust z = 11.5)) [E7:8e4ff66b].

• Deception evidence 10_scanner_cowrie.json: Honeypot connection from 62.149.181.123:30234 at 2026-08-26 09:15 (Interaction with decoy service (no legitimate use)) [E10:8ada32f0].

• Media evidence 14_voicemail.wav: Audio 14_voicemail.wav: possible editing detected (Processed with audio software: Audacity 3.4.2; 2 abrupt waveform discontinuity(ies), first at 1.4 s (possible splice)) [E14:54929452].

• Media evidence 15_cctv_clip.mp4: Video 15_cctv_clip.mp4: editing indicators found (Re-encoded or edited with Lavf62.12.101; 1 run(s) of duplicated/frozen frames (possible insertion or freeze)) [E15:4cbd925c].

Cross-domain correlation

• 02_victim_History and 03_victim_capture.pcap are linked with score 11.86 through url http://hub-cloud72.ru/dl/po_384.docm; domain hub-cloud72.ru; domain intranet.northwind.com, with related activity 0 minutes apart [E2:c36240a2][E3:cec137a2].

• 01_phishing.eml and 02_victim_History are linked with score 9.70 through url https://northwind-verify-portal.click/verify?id=65392; domain northwind-verify-portal.click; filename po_384.docm, with related activity 19 minutes apart [E1:131b6ce6][E2:c36240a2].

• 01_phishing.eml and 06_victim_disk.body are linked with score 7.62 through md5 8eded35883138246572a57a382cf77f1; filename po_384.docm, with related activity 21 minutes apart [E1:131b6ce6][E6:4a246eaa].

• 03_victim_capture.pcap and 04_victim_memory.mem are linked with score 6.96 through ip 193.214.112.115; domain hub-cloud72.ru; domain intranet.northwind.com [E3:cec137a2][E4:5307ace7].

• 02_victim_History and 04_victim_memory.mem are linked with score 5.03 through domain hub-cloud72.ru; filename po_384.docm; url https://intranet.northwind.com/home [E2:c36240a2][E4:5307ace7].

• 04_victim_memory.mem and 05_honeypot_cowrie.json are linked with score 5.00 through honeytoken decoy set Decoy credentials file on file server \\fs01\it\Passwords_backup.txt [E4:5307ace7][E5:7830e7e2]. This link is a honeytoken tripwire.

• 01_phishing.eml and 05_honeypot_cowrie.json are linked with score 4.16 through ip 23.15.47.56, with related activity 4427 minutes apart [E1:131b6ce6][E5:7830e7e2].

Deception findings

• Honeytoken 'Decoy account svc_sql78' (credential, matched on value) appears in 04_victim_memory.mem: Raw bytes at offset 438111 [E4:5307ace7].

• Honeytoken 'Decoy account svc_sql78' (credential, matched on value) appears in 05_honeypot_cowrie.json: Honeypot login SUCCESS svc_sql78/duZEitXA8c6hGd from 23.15.47.56 [E5:7830e7e2].

• Deception log 05_honeypot_cowrie.json recorded 6 events from 23.15.47.56, including 1 successful decoy logins and 0 captured downloads [E5:7830e7e2].

• Deception log 10_scanner_cowrie.json recorded 5 events from 62.149.181.123, including 0 successful decoy logins and 0 captured downloads [E10:8ada32f0].

Media forensics

• Image 07_receipt_5425.jpg has a fused manipulation score of 0.87 (threshold 0.36) and is assessed as likely manipulated [E7:8e4ff66b].

• Image 11_holiday_558.jpg has a fused manipulation score of 0.23 (threshold 0.36) and is assessed as no strong manipulation evidence [E11:49e68e2c].

Timeline

• The earliest correlated or suspicious event is at 2026-07-18 13:25: Image 07_receipt_5425.jpg: likely manipulated (score 0.87) [E7:8e4ff66b].

• The latest is at 2026-08-26 11:14: Honeypot command: id [E5:7830e7e2].

Limitations

• Findings are produced by automated modules and correlation heuristics and require investigator review before use in proceedings [E1:131b6ce6][E2:c36240a2][E3:cec137a2][E4:5307ace7][E5:7830e7e2][E6:4a246eaa][E7:8e4ff66b][E8:bfc6385b][E9:331a1c35][E10:8ada32f0][E11:49e68e2c][E12:5c9de308][E13:b83279ca][E14:54929452][E15:4cbd925c].

Correlated timeline (suspicious and pivot events)

Time (UTC)	Evidence	Severity	Event
2026-07-18 13:25:50	E7	high	Image 07_receipt_5425.jpg: likely manipulated (score 0.87)
2026-08-23 09:06:00	E2	info	Visited https://intranet.northwind.com/home
2026-08-23 09:27:00	E1	high	Email: Action required: password will expire
2026-08-23 09:27:00	E1	high	Attachment po_384.docm
2026-08-23 09:43:00	E3	high	UDP 10.8.52.183 → 1.1.1.1:53
2026-08-23 09:45:00	E3	info	DNS intranet.northwind.com
2026-08-23 09:45:00	E3	info	DNS intranet.northwind.com → 10.0.0.15
2026-08-23 09:46:00	E2	medium	Visited https://northwind-verify-portal.click/verify?id=65392
2026-08-23 09:48:00	E2	medium	Visited http://hub-cloud72.ru/dl/po_384.docm
2026-08-23 09:48:00	E2	high	Downloaded po_384.docm
2026-08-23 09:48:00	E3	info	DNS hub-cloud72.ru
2026-08-23 09:48:00	E3	info	DNS hub-cloud72.ru → 193.214.112.115
2026-08-23 09:48:00	E6	info	File modified: C:/Users/alice/Downloads/po_384.docm
2026-08-23 09:48:00	E6	info	File accessed: C:/Users/alice/Downloads/po_384.docm
2026-08-23 09:48:00	E6	info	File changed: C:/Users/alice/Downloads/po_384.docm
2026-08-23 09:48:00	E6	info	File created: C:/Users/alice/Downloads/po_384.docm
2026-08-23 09:48:02	E3	info	TCP 10.8.52.183 → 193.214.112.115:80
2026-08-23 09:48:02	E3	info	TCP 193.214.112.115 → 10.8.52.183:55361
2026-08-23 09:48:02	E3	info	HTTP GET http://hub-cloud72.ru/dl/po_384.docm
2026-08-23 09:50:59	E3	high	TCP 10.8.52.183 → 193.214.112.115:8443
2026-08-23 09:58:00	E3	high	DNS d7dmxmhkhtqxsy7elgs4gtj3w2cxpjt5fwi3w4qtu364u2qe.t.hub-cloud72.ru
2026-08-23 09:58:07	E3	high	DNS 4ucvgs54rk5dqbwvqzty35qywfbt6sj6xocggtvsjd42tyeu.t.hub-cloud72.ru
2026-08-23 09:58:14	E3	high	DNS oomso6kgm6yi35l75kdqe7aalbo5d3lml46sh77gt7235zum.t.hub-cloud72.ru
2026-08-23 09:58:21	E3	high	DNS ffarawr3jxxlgvvgfb2gfr22iut5q7sozh6vwqelricdxyy6.t.hub-cloud72.ru
2026-08-23 09:58:28	E3	high	DNS ie2e2migyddvbpbdtmpau5bnxzqwclujfo3gz4543mpmhlvq.t.hub-cloud72.ru
2026-08-23 09:58:35	E3	high	DNS jdkaqmo27nsdpsbrcw6ue4hupp7k3scgantx5oowflxzmax2.t.hub-cloud72.ru
2026-08-23 09:58:42	E3	high	DNS cone5brki7dzqtm2ojelxpqc7m7bjpaz44qspvmzwcytdvsl.t.hub-cloud72.ru
2026-08-23 09:58:49	E3	high	DNS ukzwexet4hxs3plimfgskou7amhlgqfv2ej33cfmxvtsaokg.t.hub-cloud72.ru
2026-08-26 09:15:00	E10	medium	Honeypot connection from 62.149.181.123:30234
2026-08-26 09:15:02	E10	medium	Honeypot login failed root/admin from 62.149.181.123
2026-08-26 09:15:03	E10	medium	Honeypot login failed root/123456 from 62.149.181.123
2026-08-26 09:15:04	E10	medium	Honeypot login failed admin/admin from 62.149.181.123
2026-08-26 09:15:05	E10	medium	Honeypot login failed ubnt/ubnt from 62.149.181.123
2026-08-26 11:14:00	E5	medium	Honeypot connection from 23.15.47.56:45241
2026-08-26 11:14:03	E5	medium	Honeypot login failed root/admin from 23.15.47.56
2026-08-26 11:14:09	E5	high	Honeypot login SUCCESS svc_sql78/duZEitXA8c6hGd from 23.15.47.56
2026-08-26 11:14:20	E5	medium	Honeypot command: uptime
2026-08-26 11:14:26	E5	medium	Honeypot command: ls -la

Time (UTC)	Evidence	Severity	Event
2026-08-26 11:14:32	E5	medium	Honeypot command: id

Chain of custody (hash-chained audit log)

Time (UTC)	Actor	Action	Entry hash
2026-10-30 13:09:50	investigator	case.create: Demo: northwind.com phishing intrusion (seed 42)	`d5adf248f0996779cc7a4a969d57aa9e…`
2026-10-30 13:09:50	investigator	evidence.acquire: E1 01_phishing.eml sha256=131b6ce65a66d3ca0ce4b22300fa642dc95d70ac6965f1a9f9fd983f3e1fdb7a	`19c4c0fa95783e888cde29746582061d…`
2026-10-30 13:09:50	investigator	evidence.analyse: E1 module=email status=analyzed post-analysis-hash=match	`e6b786cf8164d8ba9c1dea886382cff5…`
2026-10-30 13:09:50	investigator	evidence.acquire: E2 02_victim_History sha256=c36240a27bcc2811299f1c57f9ba0fa284377e3f178e594b1ecc70b005f527	`4713d79b7b5315b2a99e529ec9ed3b23…`
2026-10-30 13:09:50	investigator	evidence.analyse: E2 module=browser status=analyzed post-analysis-hash=match	`267e5fc77991402de2a4f9cb8afb1ac6…`
2026-10-30 13:09:50	investigator	evidence.acquire: E3 03_victim_capture.pcap sha256=cec137a294ec8cc4127e73fb3ceb0db2a4aced09e982b62efde4c3d0f	`6bdccf9282a24082069f46e79d5d54bf…`
2026-10-30 13:09:50	investigator	evidence.analyse: E3 module=network status=analyzed post-analysis-hash=match	`c7f85010755bc72fad48c3e0704b6467…`
2026-10-30 13:09:50	investigator	evidence.acquire: E4 04_victim_memory.mem sha256=5307ace7f5d487a4e0e2943022a3efe2303c28f700e30a555df311af644	`a633fa7cae77a38613ef5b35f28a4405…`
2026-10-30 13:09:50	investigator	evidence.analyse: E4 module=memory status=analyzed post-analysis-hash=match	`1bcafb599bba71b9c5ac0eb1089e89d6…`
2026-10-30 13:09:50	investigator	evidence.acquire: E5 05_honeypot_cowrie.json sha256=7830e7e2103b4b138003496b3aee066e10c1f60b9be08ef9aff74cf2	`c757457a7b3858f5ee2e765d38e06738…`
2026-10-30 13:09:50	investigator	evidence.analyse: E5 module=deception status=analyzed post-analysis-hash=match	`2cab29a815b1ed7b998c8d259f065e4e…`
2026-10-30 13:09:50	investigator	evidence.acquire: E6 06_victim_disk.body sha256=4a246eaa73687fd1994b2a222ae5aa6ae265b40e7e8d701af4d711ea8bec	`cdd57580ce0706bb02927ec11a85b371…`
2026-10-30 13:09:50	investigator	evidence.analyse: E6 module=disk status=analyzed post-analysis-hash=match	`bba2618f2b856946fd9ab10de60839fb…`
2026-10-30 13:09:50	investigator	evidence.acquire: E7 07_receipt_5425.jpg sha256=8e4ff66b53006f0208848516de97c961578adf60198ee8da7b6bb6f462e8	`f1e47991d640a2e0d1d5f1684cdf22e5…`
2026-10-30 13:09:50	investigator	evidence.analyse: E7 module=media status=analyzed post-analysis-hash=match	`e6eb9e9202e2f1b0b2563a5374f387f7…`
2026-10-30 13:09:50	investigator	evidence.acquire: E8 08_newsletter.eml sha256=bfc6385bbc02226317d05da25d839c7a84292d7da1324779c19ca42144435b	`62dede65a22bd129a312635d1f5154ff…`
2026-10-30 13:09:50	investigator	evidence.analyse: E8 module=email status=analyzed post-analysis-hash=match	`6c5be35acfdae0f005d3087a5eb54132…`
2026-10-30 13:09:50	investigator	evidence.acquire: E9 09_internal_memo.eml sha256=331a1c351f399a5233d9d3062b0b3af335b9108ad65310f277b3e254aaa	`d8b3569dd3e59b4b777968172587b7dd…`
2026-10-30 13:09:50	investigator	evidence.analyse: E9 module=email status=analyzed post-analysis-hash=match	`d54c21094282a955fd8a6e70d784903a…`
2026-10-30 13:09:50	investigator	evidence.acquire: E10 10_scanner_cowrie.json sha256=8ada32f054c69334067faf210cdf4319efd7ad5dee8afe4771df92d3	`b4fe890d6a16bb8537e9379d5e2d2509…`
2026-10-30 13:09:50	investigator	evidence.analyse: E10 module=deception status=analyzed post-analysis-hash=match	`53f38889358b73f7f595832555db474f…`
2026-10-30 13:09:50	investigator	evidence.acquire: E11 11_holiday_558.jpg sha256=49e68e2cb3770c01365f727214ddb122637e55601247e4bc1d6745305078	`384a19f5f8361eddc68b0d3f5c1064a2…`
2026-10-30 13:09:50	investigator	evidence.analyse: E11 module=media status=analyzed post-analysis-hash=match	`fdcbded28d51bbbc13fa9e4f2efa0915…`
2026-10-30 13:09:50	investigator	evidence.acquire: E12 12_colleague_History sha256=5c9de308a737e2b7ef9b3f8589341a1d3c8c75e28051b6f65a112692cb	`81ce97fac973095da08a98fdc0b40ae6…`
2026-10-30 13:09:50	investigator	evidence.analyse: E12 module=browser status=analyzed post-analysis-hash=match	`b16b4ffa31b0af51fd97f17dac1393d8…`

Time (UTC)	Actor	Action	Entry hash
2026-10-30 13:09:50	investigator	evidence.acquire: E13 13_colleague_capture.pcap sha256=b83279cac76f3e6a969482784761bc32b7337350174f0a54b824f	`6fc923780895003919 05bf2d41ce7f25…`
2026-10-30 13:09:50	investigator	evidence.analyse: E13 module=network status=analyzed post-analysis-hash=match	`4c8f99bc1c6ae1e0780be89e7ebc2452…`
2026-10-30 13:09:50	investigator	evidence.acquire: E14 14_voicemail.wav sha256=5492945241eab610ec92fb2742d0106e73cbf6a7746145d960314af1cee2f1	`55b7a8a744c1453d0855a716a5184ce5…`
2026-10-30 13:09:50	investigator	evidence.analyse: E14 module=media status=analyzed post-analysis-hash=match	`9ffa4b692f1659fb85261ea0c5a3bd99…`
2026-10-30 13:09:50	investigator	evidence.acquire: E15 15_cctv_clip.mp4 sha256=4cbd925c2c5b0219ddc401181d8a2af24565511d06397f7ea5b2d65489bcdf	`8780885b80d74c4691c9fe2f0ca0945a…`
2026-10-30 13:09:51	investigator	evidence.analyse: E15 module=media status=analyzed post-analysis-hash=match	`af741abef800795e6139c4cb863dadd5…`
2026-10-30 13:09:51	investigator	report.generate: template traceability=1.0	`2758fd06e8b83852245b9d3d0c498fd7…`